

MOBILES ARE BANNED

USN:

DEPARTMENT OF MATHEMATICS

Sub Code:	MAC21	Date:	13.06.2024	Sub:	Numerical Techniques and Differential Equations	Test:	02
Semester:	II	Time:	9.30-10.30	Term:	01-03-2024 To 22-06-2024	Marks:	30

Answer any two full questions.

O.No.	Questions	CO's	Blooms level	Marks															
1	a. If $D = \frac{d}{dx}$ and $X = X(x)$, then prove that $\frac{1}{D} X = \int X(x) dx$.	CO3	L1	2															
	b. Find the missing terms from the following data <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>$f(x)$</td><td>1</td><td>3</td><td>9</td><td>-</td><td>81</td></tr></table>	x	0	1	2	3	4	$f(x)$	1	3	9	-	81	CO4	L2	3			
	x	0	1	2	3	4													
	$f(x)$	1	3	9	-	81													
c. Solve $x^2 y'' - 4xy' + 6y = \cos[2 \log x]$	CO3	L3	5																
d. The following table gives corresponding values of pressure P and specific volume V of a superheated steam. Find the rate of change of P with respect to V when $V = 2$. <table><tr><td>V</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>P</td><td>105</td><td>42.7</td><td>25.3</td><td>16.7</td><td>13</td></tr></table>	V	2	4	6	8	10	P	105	42.7	25.3	16.7	13	CO4	L4	5				
V	2	4	6	8	10														
P	105	42.7	25.3	16.7	13														
2	a. Write the Trapezoidal rule and Simpson's $3/8^{\text{th}}$ rule for numerical integration.	CO4	L1	2															
	b. Solve $y'' + 4y' + 4y = 0$, $y(1) = 0$, $y'(0) = -1$.	CO3	L2	3															
	c. Solve $(D^2 + 4D + 5)y = x^2 e^{2x}$	CO3	L4	5															
	d. Find an interpolating polynomial representing the following data using Newton's divided difference method. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>$f(x)$</td><td>2</td><td>3</td><td>12</td><td>147</td></tr></table>	x	0	1	2	3	$f(x)$	2	3	12	147	CO4	L3	5					
x	0	1	2	3															
$f(x)$	2	3	12	147															
3.	a. What is an initial value problem and boundary value problem?	CO3	L1	2															
	b. For the following data, find x for $y = 5$, using the inverse Lagrange's method. <table><tr><td>x</td><td>2</td><td>10</td><td>17</td></tr><tr><td>y</td><td>1</td><td>3</td><td>4</td></tr></table>	x	2	10	17	y	1	3	4	CO4	L2	3							
	x	2	10	17															
	y	1	3	4															
c. Solve the differential equation $y'' + 4y = \tan 2x$ by the method of variation of parameters.	CO3	L5	5																
d. The velocity of a particle at distance from a point on its path is given by the table. Estimate the time taken to travel 60 ft by using Simpson's $1/3$ rule. <table><tr><td>s (ft)</td><td>0</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td><td>60</td></tr><tr><td>v (ft/sec)</td><td>47</td><td>58</td><td>64</td><td>65</td><td>61</td><td>52</td><td>38</td></tr></table>	s (ft)	0	10	20	30	40	50	60	v (ft/sec)	47	58	64	65	61	52	38	CO4	L3	5
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